| | Scotch Student ID # | | | |
|---|---|---|---|---|
| | | | | |
| Circle the relevant digits | 0 | 0 | 0 | 0 |
| | 1 | 1 | 1 | 1 |
| | 2 | 2 | 2 | 2 |
| | 3 | 3 | 3 | 3 |
| | 4 | 4 | 4 | 4 |
| | 5 | 5 | 5 | 5 |
| | 6 | 6 | 6 | 6 |
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| | 8 | 8 | 8 | 8 |
| | 9 | 9 | 9 | 9 |

| Teacher's Name |
|---|
| |

# Scotch College

# CHEMISTRY

# MARKING SCHEME

## SAC 1 – FUELS, GASES & THERMOCHEMISTRY

**Date: Thursday, March 17th**

| | |
|---|---|
| **Reading Time** | 0 minutes |
| **Writing Time** | 45 minutes |

| Task Sections | Marks | Your Marks |
|---|---|---|
| **Total Marks** | 38 | |

**General Instructions**

- Write your student number in the space provided above on this page.
- Please complete all questions in the spaces provided.

**Allowed Materials**

- A VCAA Chemistry Data Book has been provided.
- Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.

**At the end of the task…**

- Please ensure your student number (not name) has been filled in at the top of the page and that you have written your teacher's name.

**Electronic Devices**

Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

***Q1.*** *(1 mark)*

What is the enthalpy change, **in J**, when 5 kg of water is heated from 10°C to 18°C?

**A.** $5 \times 4.18 \times 8$

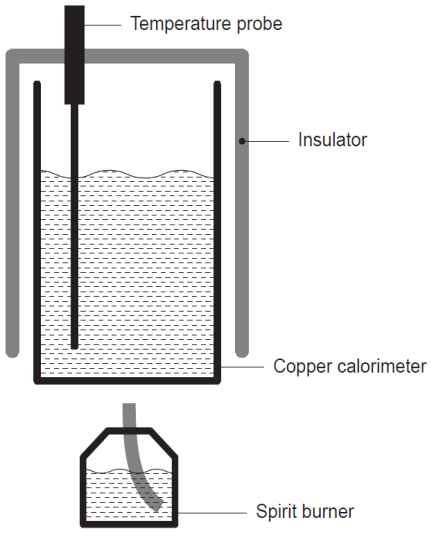
**B.** $5 \times 10^3 \times 4.18 \times 8$

**C.** $5 \times 4.18 \times (273 + 8)$

**D.** $5 \times 10^3 \times 4.18 \times (273 + 8)$

***Q2.*** (*1 mark)*

The enthalpy of combustion of a fuel was determined using the calorimeter shown.




The final result was **lower** than the literature value. Which factors could have contributed to this error?

I. Not all heat from the combustion was transferred to the calorimeter.
II. Incomplete combustion occurred.
III. The temperature probe touched the bottom of the calorimeter.

**A.** I and II only

**B.** I and III only

**C.** II and III only

**D.** I, II and III

***Q3.*** *(1 mark)*

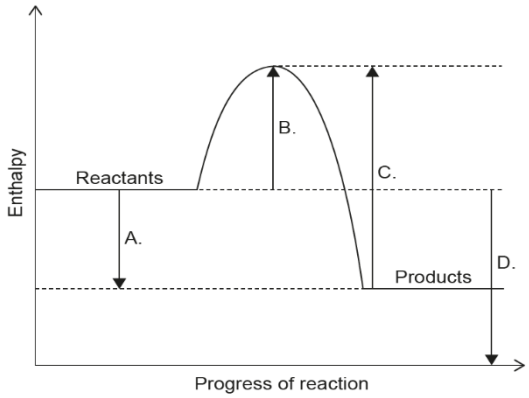
On the energy profile shown below, which letter (A-D) represents the enthalpy change of reaction, Δ*H*?

**A.**

**B.**

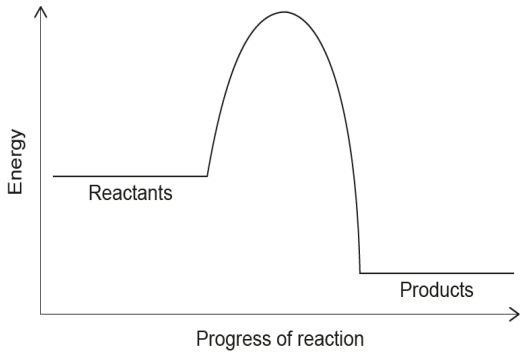
**C.**

**D.**




***Q4.*** *(1 mark)*

The potential energy profile of a reaction is shown.

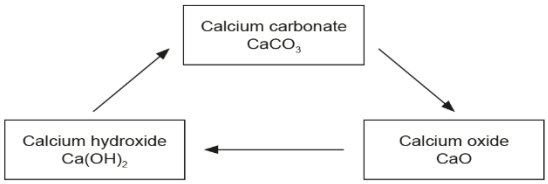



What can be determined about stability and energy change from the potential energy profile shown?

| | More Stable | Reaction |
|---|---|---|
| **A.** | Reactants | Exothermic |
| **B.** | Reactants | Endothermic |
| **C.** | Products | Exothermic |
| **D.** | Products | Endothermic |

***Q5.*** *(8 marks)*

Limestone can be converted into a variety of useful commercial products through the lime cycle. Limestone contains high percentages of calcium carbonate, $CaCO_3$, $M(CaCO_3) = 100.1$ g mol$^{-1}$.




**a.** Calcium carbonate is heated to produce calcium oxide, CaO. (3 marks)

$$CaCO_3\,(s) \rightarrow CaO\,(s) + CO_2\,(g) \quad \Delta H = 177.8 \text{ kJ mol}^{-1}$$

Calculate the volume of carbon dioxide produced at 150 kPa and 100 °C when 555 g of calcium carbonate decomposes.

$n(CaCO_3) = 555/\ 100.1 \quad = 5.50$ mol *

$V(CaCO_3) = (5.50 \times 8.31 \times 373) / 150 \quad = 115$ L * (114 also accepted) -1 if not 3 sig figs

**b.** Calcium hydroxide reacts with carbon dioxide to reform calcium carbonate. (3 marks)

$$Ca(OH)_2\,(aq) + CO_2\,(g) \rightarrow CaCO_3\,(s) + H_2O\,(l) \quad \Delta H = -69.1 \text{ kJ mol}^{-1}$$

Determine the mass, in g, of $CaCO_3$ (s) produced by reacting 2.41 L of $2.33 \times 10^{-2}$ M of $Ca(OH)_2$ (aq) with 0.750 L of $CO_2$ (g) at SLC.

$n(Ca(OH)_2) = 2.41 \times (2.33 \times 10^{-2}) = 5.62 \times 10^{-2}$ mol *

$n(CO_2) = 0.750/\ 24.8 \quad = 3.02 \times 10^{-2}$ mol* (LIMITER)

$n(CaCO_3) = n(CO_2)$

$m(CaCO_3) = 3.02 \times 10^{-2} \times 100.1 \quad = 3.03$ g *

**c.** Given the following thermochemical equations, determine the thermochemical equation for the formation of calcium hydroxide ($Ca(OH)_2$) from calcium oxide (CaO) and water. (2 marks)

$$Ca(OH)_2\,(aq) + CO_2\,(g) \rightarrow CaCO_3\,(s) + H_2O\,(l) \qquad \Delta H = -69.1 \text{ kJ mol}^{-1}$$

$$CaCO_3\,(s) \rightarrow CaO\,(s) + CO_2\,(g) \qquad \Delta H = +177.8 \text{ kJ mol}^{-1}$$

$$CaO\,(s) + H_2O\,(l) \rightarrow Ca(OH)_2\,(aq)^* \quad \Delta H = -108.7 \text{ kJ mol}^{-1} *$$

Comment on missing states, but mark not to be deducted.

***Q6.*** *(12 marks)*

**a.** Ethanol is a biofuel that can be mixed with petrol.

Write the **thermochemical** equation for the complete combustion of liquid ethanol. (2 marks)

$C_2H_5OH$ (l) + $3O_2$ (g) → $3H_2O$ (l) + $2CO_2$ (g) $\Delta H = -1360$ kJ $mol^{-1}$

* States (water must be l) and correct equation, * enthalpy

**b.** All fuels produce carbon dioxide during combustion. Explain, including a suitable equation, why bioethanol can be considered to be more carbon neutral than ethanol derived from fossil fuels. (3 marks)

$CO_2$ consumed as the plant used to make the biofuel grows, is equal to the $CO_2$ produced during fermentation and combustion*, therefore net zero emissions. *

* $6H_2O + 6CO_2 \rightarrow C_6H_{12}O_6 + 6O_2$

**c.** Biodiesel can be made when ethanol reacts with vegetable oil. What other product is formed in this process? (1 mark)

Glycerol * (must be correctly spelled)

Methyl palmitate, $C_{17}H_{34}O_2$, is a component of a type of biodiesel. It is liquid at room temperature. The molar enthalpy of combustion was determined using a bomb calorimeter. The calorimeter was calibrated by passing 3.40 amps and 5.61 volts through an electric heater for 240 seconds. The temperature of the calorimeter rose by 1.75 °C.

**d.** Calculate the calibration factor of the calorimeter. (3 marks)

CF = VIt/ ΔT

= (3.40 x 5.61 x 240)*/ 1.75

=2.62 x$10^3$ * J $°C^{-1}$ or 2.62 kJ $°C^{-1}$ *correct unit

A 0.578 g sample of the methyl palmitate was then burned in excess oxygen in the calorimeter and the temperature rose by 6.19 °C. M($C_{17}H_{34}O_2$) = 270.0 g $mol^{-1}$.

**e.** Calculate the amount of energy, in kJ, absorbed by the calorimeter when the sample of methyl palmitate was burned. (1 mark)

q = CF x ΔT

= 2.62 x 6.19

= 16.2* kJ (Note: When marking this question, the 2.62 (rounded to correct significant figures) needed to be used to give the final answer. Leaving it in your calculator from part d is marked incorrect by VCAA).

**f.** Calculate the amount of energy released, in kJ, by the combustion of 1.00 mol of methyl palmitate. (2 marks)

$n(C_{17}H_{34}O_2)$ = 0.578/ 270

= $2.14 \times 10^{-3}$ mol *

q/mol = 16.2/ $2.14 \times 10^{-3}$

= $7.57 \times 10^{3}$ kJ*

***Q7.*** *(12 marks)*

Vegetable oils require conversion to biodiesel for use in current internal combustion engines.

**a.** A student obtained a sample of petrodiesel and biodiesel for the purpose of investigating their properties. Unfortunately, when they placed the samples into iced water, the labels came off. Use the results below to identify the samples. Fill in the name of tube A below. (1 mark)

| | Flow rate (L $min^{-1}$) | |
|---|---|---|
| Temperature (°C) | Tube A | Tube B |
| 0 | Crystalised | 0.11 |
| Room temp | 0.16 | 0.25 |
| 30 | 0.37 | 0.55 |

Tube A __Biodiesel *_____

**b.** What structural difference between petrodiesel and biodiesel are most likely to be responsible for the property named in 7a. Explain your answer by describing the structure of both fuels.

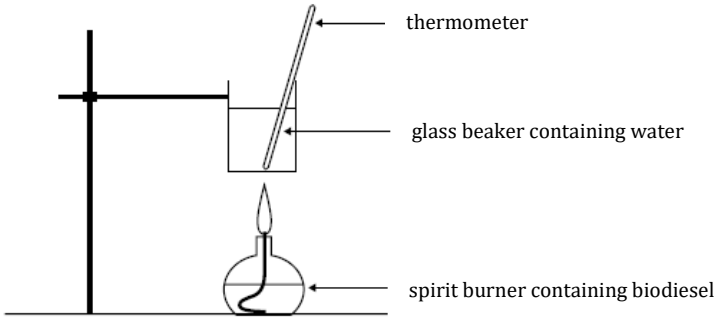
(3 marks)

Petrodiesel is a non-polar hydrocarbon chain. *

Biodiesel has a non-polar hydrocarbon chain with a polar ester group on one end. *

The ester group allows the biodiesel to form dipole-dipole bonds to itself which are stronger than the petrodiesel dispersion forces. * (MUST mention both dipole-dipole and dispersion)

Having tried to determine the energy content of biodiesel by burning it in a crucible sitting in water and being disappointed by the amount of heat lost, the student poured the remaining biodiesel into a spirit burner and tried again.




The results are shown below.

| | |
|---|---|
| Mass of water (g) | 198.203 |
| Mass of biodiesel burnt (g) | 2.076 |
| Initial temperature of water (°C) | 19.2 |
| Final temperature of water (°C) | 25.8 |

**c.** Calculate the energy, in Joules, absorbed by the water. (2 marks)

q = 198.203 x 4.18 x (6.6 *)

= $5.5 \times 10^3$ J * Comment if incorrect sig figs

**d.** Determine the heat of combustion of the biodiesel in kJ $g^{-1}$. (1 mark)

Hc = 5.5 * / 2.076

= 2.6 kJ $g^{-1}$ * Comment incorrect sig figs

**e.** The results were still lower than the value listed in the text book, so the student decided to make two changes to the set up shown above. What are two changes that would **reduce energy loss** in this apparatus? (2 marks)

** Any two of: insulation shield, lid on the beaker, use metal instead of glass

**f.** While packing the equipment away the student noticed that the bottom of the glass beaker had turned black (covered in carbon soot). Explain this observation by writing a balanced chemical equation using the fuel octane, $C_8H_{18}$. States are not required. (1 mark)

$C_8H_{18} + 9/2\ O_2 \rightarrow 9H_2O + 8C$ * or $2C_8H_{18} + 9O_2 \rightarrow 18H_2O + 16C$

***Q8.*** *(4 marks)*

Pig and cattle farmers across Australia are using the biogas from their animal's waste to produce the electricity for their farms. An average pig will produce 2.98 kg of methane gas per day.

**a.** Calculate the maximum energy, in kJ, that could be generated by combustion of the methane produced by an average pig in one year (365 days). (1 mark)

$Hc(CH_4) = 55.6$ kJ $g^{-1}$ (Data Book)

Energy in 1 year = $55.6 \times (2.98 \times 10^3) \times 365$ = $6.05 \times 10^7$ * kJ

**b.** The methane gas is collected in tanks before being used in a combustion engine to produce electricity. A farmer setting up a biogas collection system of flexible polymer tanks. What volume, in ML, would they expect to collect from 200 pigs, ***per day***, if the gas is stored at SLC? (2 marks)

$n(CH_4) = (2.98 \times 10^3 \times 200)/ 16.0$ = $3.72 \times 10^4$ mol *

$V(CH_4) = 3.72 \times 10^4 \times 24.8 = 9.24 \times 10^5$ L = 0.924 ML *

**c.** While considerably more expensive, some farmers store their methane by increasing the pressure to 4550 kPa. If they keep the temperature at 25.0 °C, what volume, in ML, would the volume from question **8b** (above) of methane occupy? (1 mark)

$100 \times 0.924 = 4550 \times V_2$ $V_2 = 2.03 \times 10^{-2}$ * ML

(Must use correctly rounded figure from part b in calculation for part c)

**End of SAC**